

## PROFESSIONAL EXPERIENCE

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- **Meta, Inc.** Menlo Park, US  
*Senior Machine Learning Engineer @ Instagram-Threads Feed Relevance* Mar. 2026 – Present
  - Support new and low signal users for Threads growth, optimizing across the stack (retrieval, ranking model, value model, re-rank, cold-start).
  - Serve as technical lead for the re-ranking layer, developing end-to-end interest-matching and diversity solutions to improve relevance and user experience.
- **TikTok, Inc.** San Jose, US  
*Senior Machine Learning Engineer @ TikTok Data-Plus* Jan. 2025 – Mar. 2026
  - Led end-to-end optimization of TikTok content push recommendation, spanning **approximate-nearest-neighbor (ANN) retrieval, cold-start embedding models, multi-stage ranking, and LLM-based creative optimization**.
  - Shipped **20+ algorithm launches** at US scale, improving personalization quality and driving **+0.4% DAU** (~1M users) incremental **user growth**.
  - Mentored **10+ junior engineers and interns** across Shanghai and US teams and helped to scaled the team from 2 to 20.
- **ByteDance (TikTok)** Shanghai, CN  
*Senior Machine Learning Engineer @ TikTok Data* May. 2022 – Jan. 2025
  - Founding engineer and technical lead of the social recommendation modeling. Helped to scale TikTok's **friends recommendation system (people you may know)** from 0 to 1;
  - Shipped 30+ launches per year across **offline/online retrieval, end-to-end ranking, and rec sys strategies**. Contributed 80%+ of our group goal per year. Achieved **300% growth in business metrics** and contributed **~1%+ DAU lift**, collaborating cross-functionally with backend, infra, and product teams.
  - Proposed and implemented a **real-world-social network project**, introducing relationship feedback loops and hybrid retrieval mechanisms. Led a collaboration team with over **20 people**, resulting in **15+** impactful launches and **0.5% dau lift**.
- **Tencent, Inc.** Shenzhen, CN  
*Machine Learning Researcher @ IEG Lightspeed&Quantum Studios (LOL Mobile)* Apr. 2021 – May. 2022
  - Developed **new-player skill detection system** for League of Legends Mobile; iterated 10+ model versions across client/server, reducing one-sided matches by **20%** and improving retention by **+1% (30M DAU)**.
  - Built a **transformer-based matchmaking model with multi-task learning**, predicting game quality and fairness; increased player satisfaction by **10%**, reduced one-sided games by **20%**, and improved retention by **1.5%**
  - Delivered **data infra tools** (Spark/Go/K8s dashboards, user profiling services), resolving 50+ tickets and supporting multiple game features. Awarded **company-level business breakthrough award**.

## EDUCATION

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- **University of California, San Diego** San Diego, US  
*Master of Science in Electrical and Computer Engineering* Sept. 2018 – Dec. 2020
- **Northwestern Polytechnical University** Xi'an, CN  
*Bachelor of Science in Aerospace Engineering* Sept. 2014 – Jun. 2018

## PUBLICATIONS

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- **Kai Wang**, Ge Fan, Chaoyun Zhang, Yuyang Jiang, and Yuze Liu. CHAMP: Cross-domain Hybrid Architecture for Matchmaking and Prediction in Online Multi-Player Games. *CIKM*, 2026.
- **Kai Wang**, Chaoyun Zhang, Hao Chen, Ge Fan, Yingjie Li, Lifang Wu, and Bingchao Zheng. Quickskill: Novice skill estimation in online multiplayer games. *CIKM*, 2022.
- **Kai Wang**, Chunxu Shen, Chaoyun Zhang, and Wenye Ma. AdnFM: An attentive densenet based factorization machine for CTR prediction. *ICCDE*, 2022. **Best Oral Presentation Award**.

- Ge Fan, Chaoyun Zhang, and **Kai Wang**. Cupid: Improving battle fairness and position satisfaction in online MOBA games with a re-matchmaking system. *CSCW*, 2024.
- Ge Fan, Chaoyun Zhang, **Kai Wang**, and Junyang Chen. Mv-han: A hybrid attentive networks based multi-view learning model for large-scale content recommendation. *ASE*, 2022.

## PROFESSIONAL SERVICES

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- **PC** CIKM 2024, CIKM 2025, CIKM 2026

## TECHNICAL SKILLS

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**Machine Learning:** Recommender systems, retrieval and ranking, deep learning, representation learning, cold start, LLM applications, A/B testing

**Languages/Tools:** Python, Scala, SQL, Go, C++, PyTorch, TensorFlow, Spark, Hadoop, Docker, Kubernetes